CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 12: Reflection

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Reflection of figures

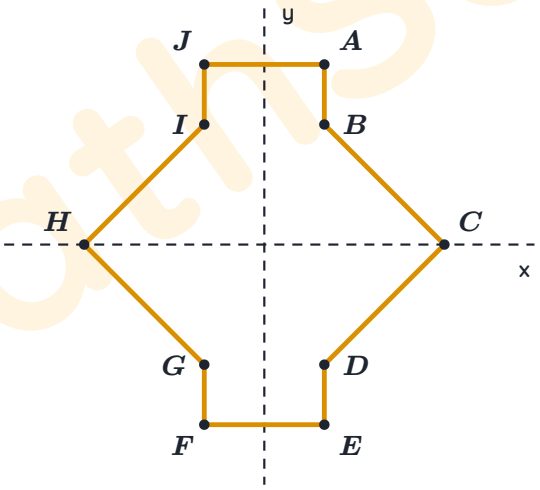
1. Plot $A(1,3)$, $B(1,2)$ and $C(3,0)$ on graph paper. (a) Reflect A and B in the x-axis to get E and D. (b) Reflect A and B in the origin to get F and G. (c) Reflect A, B and C in the y-axis to get J, I and H. (d) Join $A, B, C, D, E, F, G, H, I, J$ in order and name the closed figure. ICSE 2025, 5 marks

SOLUTION

(a) Reflection in the x-axis changes the sign of y: $E(1,-3)$ and $D(1,-2)$.

(b) Reflection in the origin changes both signs: $F(-1,-3)$ and $G(-1,-2)$.

(c) Reflection in the y-axis changes the sign of x: $J(-1,3)$, $I(-1,2)$ and $H(-3,0)$.



(d) The figure has 10 sides and two of its angles point inwards on each side.

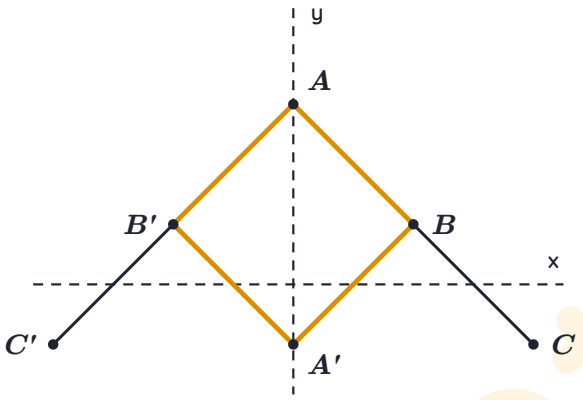
Answer: a concave decagon, shaped like a double-headed arrow

2. Plot $A(0,3)$, $B(2,1)$ and $C(4,-1)$. (a) Reflect B and C in the y-axis to get B' and C', and write their coordinates. (b) Reflect A in the line BB' to get A', and write its coordinates. (c) Join A, B, A', B' in order and name the figure. ICSE 2024, 5 marks

SOLUTION

(a) Reflection in the y-axis changes the sign of x: $B'(-2,1)$ and $C'(-4,-1)$.

(b) BB' is the horizontal line $y=1$. A is 2 units above it, so A' is 2 units below it: $A'(0,-1)$.



(c) $AB = BA' = A'B' = B'A = \sqrt{2^2+2^2} = \sqrt{8}$, and the diagonals $AA' = BB' = 4$ are equal.

Answer: $B'(-2,1)$, $C'(-4,-1)$, $A'(0,-1)$; $ABA'B'$ **is a square**

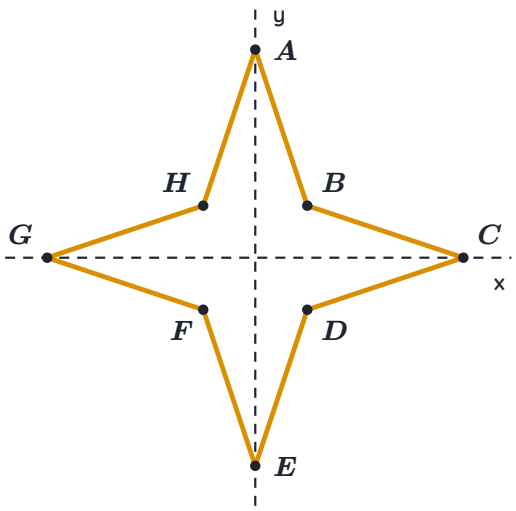
3. Plot $A(0,4)$, $B(1,1)$ and $C(4,0)$. (a) Reflect A and B in the x-axis to get E and D. (b) Reflect B in the origin to get F, and write its coordinates. (c) Reflect B and C in the y-axis to get H and G. (d) Join $A, B, C, D, E, F, G, H, A$ in order and name the closed figure. ICSE 2023, 5 marks

SOLUTION

(a) $E(0,-4)$ and $D(1,-1)$.

(b) Reflection in the origin changes both signs: $F(-1,-1)$.

(c) $H(-1,1)$ and $G(-4,0)$.



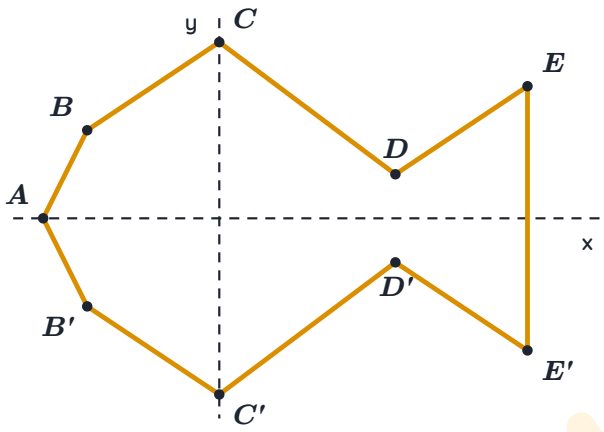
(d) The figure has 8 sides, with four of its corners pointing inwards.

Answer: $F(-1,-1)$**; the figure is a concave octagon in the shape of a four-pointed star**

4. Plot $A(-4,0)$, $B(-3,2)$, $C(0,4)$, $D(4,1)$ and $E(7,3)$. Reflect B, C, D and E in the x-axis to get B', C', D' and E'. Join $A, B, C, D, E, E', D', C', B', A$ in order and name the closed figure. ICSE 2020, 4 marks

SOLUTION

Reflection in the x-axis changes the sign of y: $B'(-3,-2)$, $C'(0,-4)$, $D'(4,-1)$ and $E'(7,-3)$.

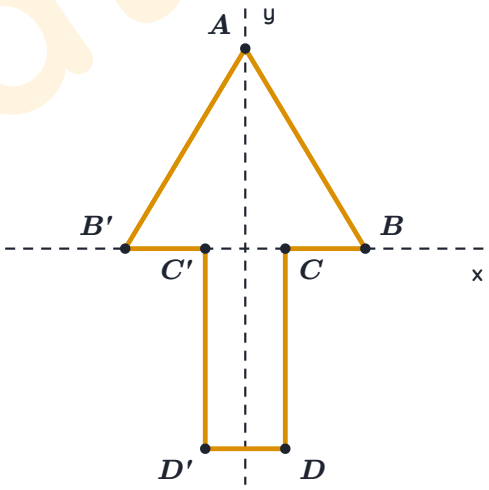


Answer: the figure is shaped like a fish (a concave nonagon, symmetric about the x-axis)

5. Plot $A(0,5)$, $B(3,0)$, $C(1,0)$ and $D(1,-5)$. Reflect B, C and D in the y-axis to get B', C' and D', and write their coordinates. Join $A, B, C, D, D', C', B', A$ in order and name the closed figure.
ICSE 2019, 4 marks

SOLUTION

Reflection in the y-axis changes the sign of x: $B'(-3,0)$, $C'(-1,0)$ and $D'(-1,-5)$.

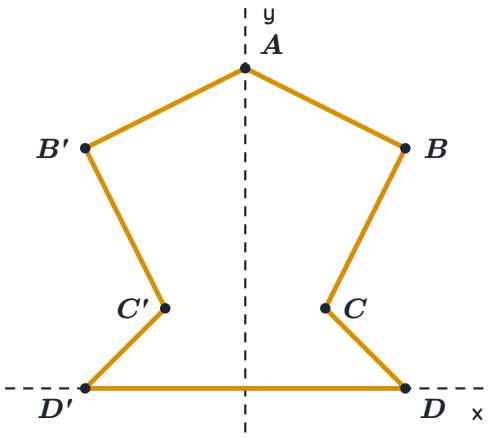


Answer: $B'(-3,0)$, $C'(-1,0)$, $D'(-1,-5)$; the figure is an arrow head (a concave heptagon)

6. Plot $A(0,4)$, $B(2,3)$, $C(1,1)$ and $D(2,0)$. Reflect B, C and D in the y-axis to get B', C' and D', and write their coordinates. Join $A, B, C, D, D', C', B', A$ in order and write the equation of the line of symmetry of the figure. ICSE 2017, 5 marks

SOLUTION

Reflection in the y-axis changes the sign of x: $B'(-2,3)$, $C'(-1,1)$ and $D'(-2,0)$.



The figure is its own mirror image in the y-axis.

Answer: $B'(-2,3)$, $C'(-1,1)$, $D'(-2,0)$; the line of symmetry is $x = 0$

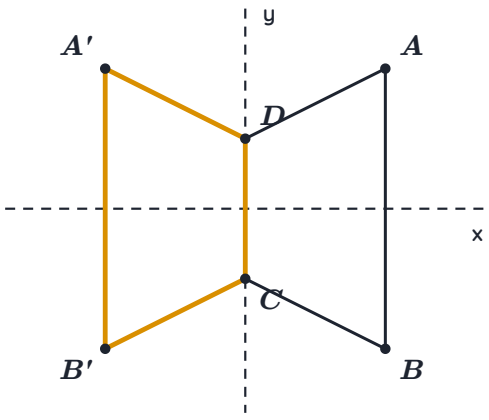
2 Invariant points

7. $ABCD$ is a quadrilateral with vertices $A(2,2)$, $B(2,-2)$, $C(0,-1)$ and $D(0,1)$. (a) Reflect $ABCD$ in the y-axis to get $A'B'CD$, and write the coordinates of A' and B'. (b) Name two points that are invariant under this reflection. (c) Name the polygon $A'B'CD$. ICSE 2018, 4 marks

SOLUTION

(a) $A'(-2,2)$ and $B'(-2,-2)$.

(b) C and D lie on the y-axis, which is the mirror line, so they are invariant.



(c) $A'B' \parallel DC$ (both vertical) and $A'D = B'C = \sqrt{5}$.

Answer: $A'(-2,2)$, $B'(-2,-2)$; invariant points C and D; $A'B'CD$ is an isosceles trapezium

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu